LAB 6

CODE AND BRANCH COVERAGE ANALYSIS

Electricity Bill Application

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Course / Subject	Software Engineering Lab
Experiment	Lab 6 – Code and Branch Coverage Analysis

Academic Lab Report

1. Introduction

Code coverage is a software testing metric that indicates how much of a program is executed when a test suite runs. This experiment uses Python's coverage.py tool together with pytest to measure statement (line) coverage and branch coverage for an electricity bill application.

The application calculates electricity energy charges based on consumption slabs and connection type, applies late-payment fees, and calculates eligible subsidies.

2. Objectives

- Understand the difference between statement coverage and branch coverage.
- Run the supplied tests and record the initial coverage.
- Develop additional test cases for normal, boundary, and invalid-input scenarios.
- Use coverage.py to measure the final statement and branch coverage.
- Achieve at least 90% coverage, as required by the lab handout.

3. Software Requirements

Component	Version / Details
Operating system	Windows
Programming language	Python 3.14.6
Testing framework	pytest 9.1.1
Coverage tool	coverage.py 7.17.0

4. Application Description

The module electricity_bill.py contains three functions and a fixed-charge mapping:

energy_charge(units, connection): Validates the connection type and non-negative units; calculates charges using the 0–100, 101–300, and above-300 unit slabs; doubles the energy charge for commercial connections; and adds the connection's fixed charge.

late_fee(bill_amount, days_late): Returns zero when payment is not late, applies a ₹50 fee for up to 15 days late, and a ₹100 fee after 15 days. For bills above ₹2,000 and more than 15 days late, it adds bill_amount // 20.

subsidy(units, is_senior, has_solar): Provides a ₹50 senior-citizen subsidy for consumption up to 100 units, adds a ₹100 solar subsidy, and sets the subsidy to zero when solar is present and consumption exceeds 300 units.

5. Testing Approach

The supplied starter tests were first executed to establish a baseline. Additional tests were then written in two separate files, as required by the lab deliverables:

- CodeCoverageTest_PES1UG24CS381_application.py – 12 tests targeting execution of the application's statements.
- BranchCoverageTest_PES1UG24CS381_application.py – 19 tests targeting both outcomes of conditional decisions and important boundary values.

6. Test Scenarios

Function	Scenarios exercised
Energy charge	Domestic and commercial connections; low, middle, and high consumption slabs; invalid connection; negative units.
Late fee	On-time payment; 1–15 days late; more than 15 days late; bill amount above and equal to ₹2,000.
Subsidy	Eligible senior citizen; ineligible consumption; non-senior customer; solar and non-solar cases; exactly 300 and above 300 units.

7. Commands Used

```
python -m pytest -v

coverage erase

coverage run -m pytest CodeCoverageTest_PES1UG24CS381_application.py -v

coverage report -m electricity_bill.py

coverage erase

coverage run --branch -m pytest BranchCoverageTest_PES1UG24CS381_application.py -v

coverage report -m electricity_bill.py

coverage html -d htmlcov_branch
```

8. Results and Coverage Analysis

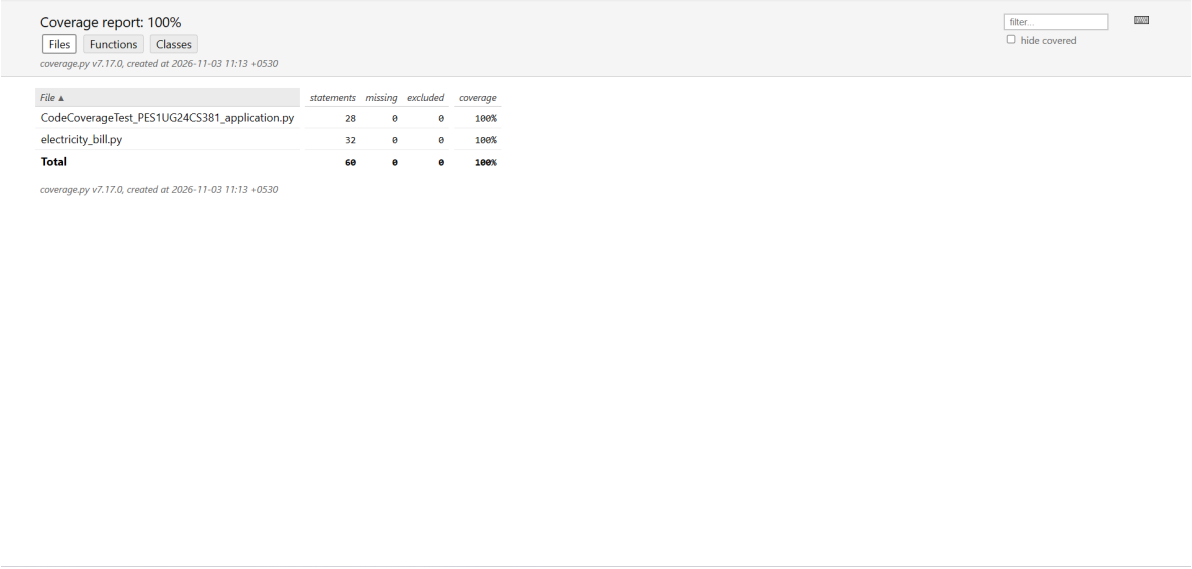
The initial supplied test suite contained two passing tests. The baseline report showed 38% statement coverage for electricity_bill.py and 31% branch coverage. After adding the dedicated test suites, the application achieved full statement and branch coverage.

Metric	Code coverage run	Branch coverage run	Result
Tests passed	12	19	All passed
Application statements	32/32	32/32	100%
Branches	Not measured in line-only run	22/22	100%
Partial branches	Not measured	0	0
Application coverage	100%	100%	Exceeds 90% requirement

Note: The coverage percentages above refer to the application module electricity_bill.py. The HTML reports can also include the test file in their overall totals; the application-only report is the relevant measure for evaluating the application code.

9. Evidence Screenshots

The following screenshots are included as evidence. The code coverage screenshots show the statement coverage report and its function/class views. The branch coverage screenshot shows statement and branch metrics, including 22 branches and zero partial branches.



Coverage report: 100%

Files | Functions | Classes

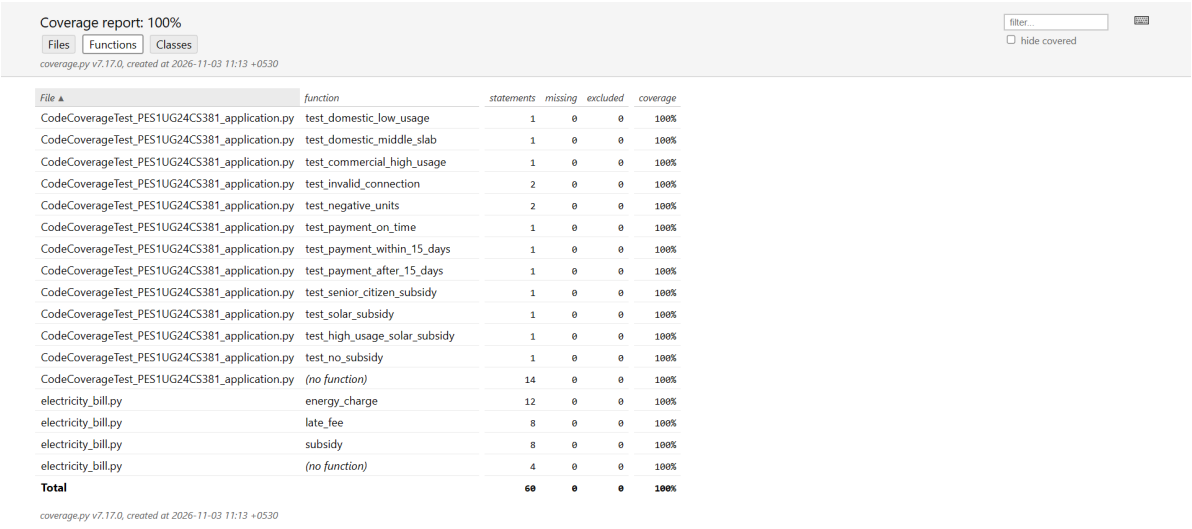
filter... hide covered

coverage.py v7.17.0, created at 2026-11-03 11:13 +0530

File ▲	statements	missing	excluded	coverage
CodeCoverageTest_PES1UG24CS381_application.py	28	0	0	100%
electricity_bill.py	32	0	0	100%
Total	**60**	**0**	**0**	**100%**

coverage.py v7.17.0, created at 2026-11-03 11:13 +0530

Figure 1. Code coverage – Files summary (100%).



Coverage report: 100%

Files | Functions | Classes

filter... hide covered

coverage.py v7.17.0, created at 2026-11-03 11:13 +0530

File ▲	function	statements	missing	excluded	coverage
CodeCoverageTest_PES1UG24CS381_application.py	test_domestic_low_usage	1	0	0	100%
CodeCoverageTest_PES1UG24CS381_application.py	test_domestic_middle_slab	1	0	0	100%
CodeCoverageTest_PES1UG24CS381_application.py	test_commercial_high_usage	1	0	0	100%
CodeCoverageTest_PES1UG24CS381_application.py	test_invalid_connection	2	0	0	100%
CodeCoverageTest_PES1UG24CS381_application.py	test_negative_units	2	0	0	100%
CodeCoverageTest_PES1UG24CS381_application.py	test_payment_on_time	1	0	0	100%
CodeCoverageTest_PES1UG24CS381_application.py	test_payment_within_15_days	1	0	0	100%
CodeCoverageTest_PES1UG24CS381_application.py	test_payment_after_15_days	1	0	0	100%
CodeCoverageTest_PES1UG24CS381_application.py	test_senior_citizen_subsidy	1	0	0	100%
CodeCoverageTest_PES1UG24CS381_application.py	test_solar_subsidy	1	0	0	100%
CodeCoverageTest_PES1UG24CS381_application.py	test_high_usage_solar_subsidy	1	0	0	100%
CodeCoverageTest_PES1UG24CS381_application.py	test_no_subsidy	1	0	0	100%
CodeCoverageTest_PES1UG24CS381_application.py	*(no function)*	14	0	0	100%
electricity_bill.py	energy_charge	12	0	0	100%
electricity_bill.py	late_fee	8	0	0	100%
electricity_bill.py	subsidy	8	0	0	100%
electricity_bill.py	*(no function)*	4	0	0	100%
Total		**60**	**0**	**0**	**100%**

coverage.py v7.17.0, created at 2026-11-03 11:13 +0530

Figure 2. Code coverage – Function-level summary (100%).

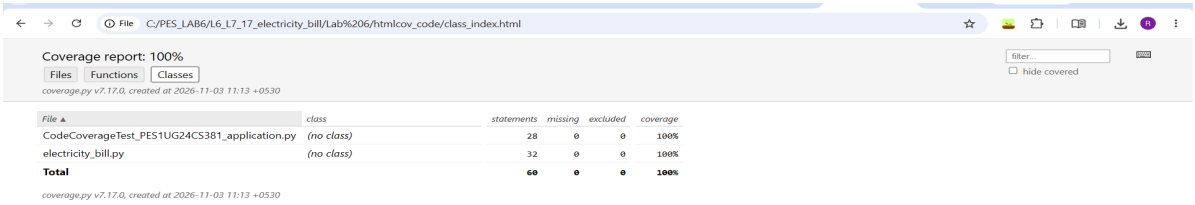


Figure 3. Code coverage – Classes view.

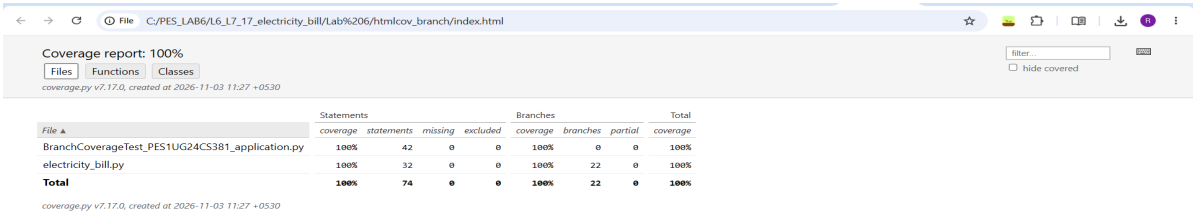


Figure 4. Branch coverage – Files summary (100%; 22 branches, 0 partial).

10. Conclusion

The experiment demonstrated how statement coverage and branch coverage provide complementary views of test completeness. The additional test cases exercised the electricity bill application's validation rules, slab boundaries, late-fee decisions, and subsidy conditions. The final results achieved 100% statement coverage and 100% branch coverage for electricity_bill.py, exceeding the minimum 90% target specified in the lab handout.

11. Submission Checklist

- Application file: electricity_bill.py
- Original starter test file: test_electricity_bill.py
- Code coverage test file: CodeCoverageTest_PES1UG24CS381_application.py
- Branch coverage test file: BranchCoverageTest_PES1UG24CS381_application.py
- Code coverage screenshot(s) and branch coverage screenshot
- This Word report
- Two separate Git commits and successful push to the submission repository